Third System
AI Presence Index

At the Boundary

AIAS v0.19 — Audiophile Headphones

May 2026 · Pablo Ulpiano González Castro

Three Iwachu cases on a third substrate family. A Recognition ceiling holding at exactly 0.500 modal share. A cultural channel waiting for v0.20.

The audiophile-headphones panel met the Recognition ceiling that v0.18 met before it. Cell A_Heritage stops at exactly 0.500 modal share — failing C2 by a single brand at the ceiling. Cell B_Boutique sits well above the threshold at 0.875. H_C3 routes to UNDETERMINED per pre-registered Rule 4. But three new Iwachu cases extend the Recognition × Recall dissociation construct to a third substrate family, and a cultural-channel asymmetry — three Type 1 brands, zero Type 2 brands — identifies a panel-composition limit that points cleanly at v0.20's substrate selection. The phase is informative below the headline.

The AIAS™ Presence Measurement Protocol v1.4 multi-component construct widens to three substrate families. The C3 ranking-coherence rescue attempted on a same-IL Heritage × Boutique design didn't land — the audiophile-headphone substrate produces Recognition ceiling effects in both cells. Three pre-registered hypotheses; one UNDETERMINED, one PARTIAL, one descriptive. The methodology gains an empirical anchor; the construct gains three more cases.

v0.19 was designed as the C3 rescue phase. v0.18 left the ranking-coherence test structurally degenerate — Cell A's flat Recognition (7 of 8 brands tied at C_P = 6/6) and Cell B's flat Recall (7 of 8 brands tied at 0 mentions) both undermined within-cell rank correlation. v0.19 selected a substrate engineered for within-cell variance in both dimensions: audiophile headphones, two same-IL cells split by audiophile headphone product-line emergence year (Heritage ≤ 2008, Boutique > 2008), 16 brands worldwide pre-floor, English-language anchored throughout.

The substrate did not deliver C3. Cell A_Heritage produced Recognition modal share = 0.500 (four of eight brands at C_P = 6/6: Sennheiser, Beyerdynamic, Grado, Audio-Technica) — exactly at the C2 strict-less-than threshold and failing by a single brand at the ceiling. Cell B_Boutique produced modal share = 0.875 (seven of eight brands at C_P = 6/6; only Spirit Torino at 5/6 below ceiling). Both cells fail C2; H_C3 routes to UNDETERMINED per pre-registered DEVIATIONS Rule 4. Per the rule, panel is NOT substituted; the substantive finding stands.

The substantive value of v0.19 sits below the UNDETERMINED headline. Three Iwachu-pattern cases (Phase A C_P 5/6 AND Phase B category-anchored mentions 2/18) replicate on the headphone substrate: ZMF Headphones (6/6, 1/18), Spirit Torino (5/6, 0/18), and Final Audio (6/6, 0/18) — all Cell B_Boutique. The v1.4 multi-component construct now has 13 cases across three substrate families (v0.17 Iwachu cross-cultural + v0.18 indie fragrance 9 cases + v0.19 audiophile 3 cases), strengthening its empirical foundation beyond what a single phase can provide. PARTIAL on the matrix, but the third-substrate-family contribution is the headline.

A cultural-channel asymmetry surfaces three Type 1 cases (Audeze, HiFiMan, Dan Clark Audio: category-anchored mentions 5/18 AND cultural-footprint mentions 2/18) and zero Type 2 cases. The Type 2 absence reflects panel composition, not the absence of the cultural-footprint Recall pathway: the Heritage × Boutique audiophile-electronics design places both cells inside the category-recognition envelope. A v0.20 substrate including a mass-consumer headphone cell (Bose, Beats, Apple, Skullcandy, JBL) would test Type 2 cleanly. The finding identifies a load-bearing v0.20 substrate selection criterion.

What we measured

Two acquisition phases scored by pre-registered decision rules. Phase A measures **Recognition** — whether the AI panel recognizes a brand as belonging to the category. Phase B measures **Recall** — whether the AI panel surfaces the brand when asked to enumerate category members. Verdicts route through a C1 → C2 → C3 cascade per the locked pre-registration.

Phase A asks each of the six reference panel models a single question per brand: ***'Is the brand X commonly recognized as a brand of audiophile headphones?'*** The brand's Recognition score (C_P, range 0..6) is the count of recognition-positive responses across the panel. Cell-level variance is tested via C2 modal share — the share of brands at the cell's most common C_P value, which must be strictly less than 0.50 for the cell to pass.

Phase B asks the same six models six queries: three category-anchored frames (***best audiophile headphones***, ***high-quality headphones for serious music listening***, ***reference-grade headphones***) that load-bear on the verdict matrix; plus three cultural-footprint frames (***most famous headphone brands***, ***most well-known among general consumers***, ***strongest cultural recognition***) that operate as descriptive sensitivity for the cultural-footprint Recall channel surfaced as a v0.18 finding. Each brand has at most 18 observation slots per channel (3 frames × 6 models).

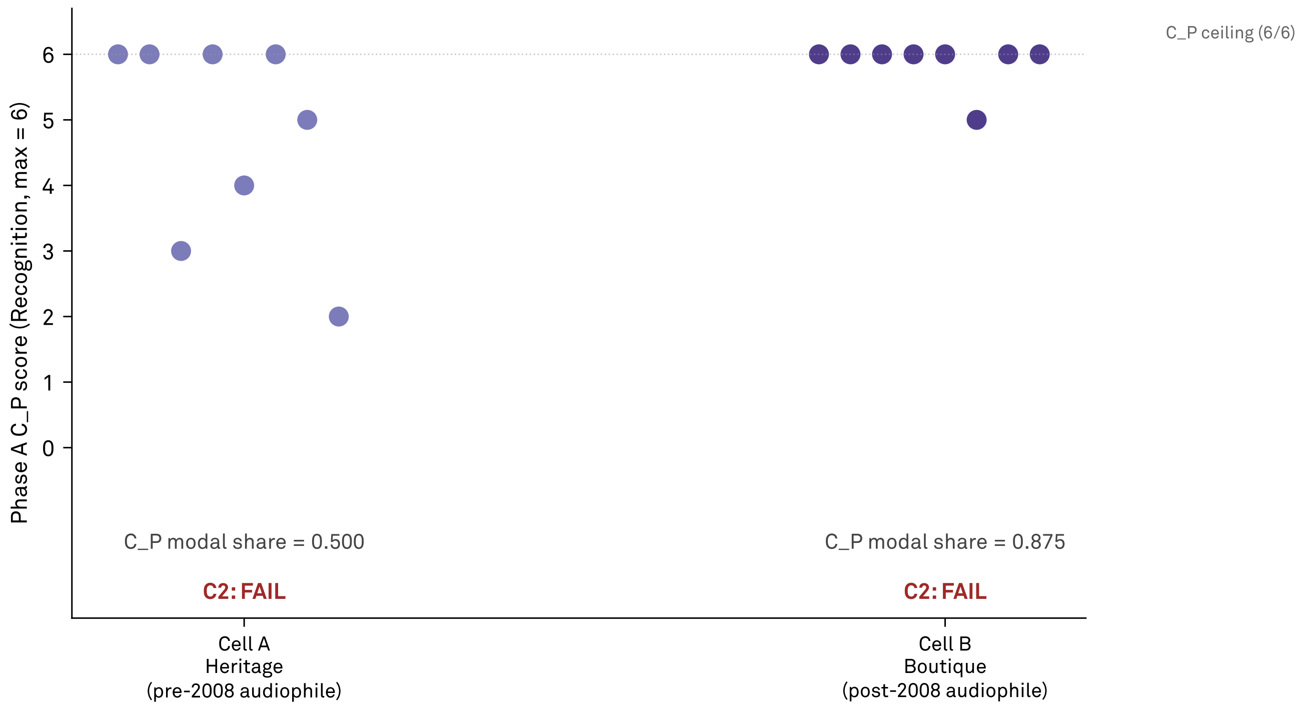
The Recognition × Recall dissociation threshold (Iwachu-pattern) requires C_P ≥ 5/6 AND category-anchored mentions ≤ 2/18 — the brand is recognized in 5+ of 6 single-brand probes, but surfaces in 2 or fewer of 18 enumeration responses. The Type 1 / Type 2 channel-asymmetry thresholds replicate this asymmetric structure across the cultural-footprint axis. The pre-registration is locked at commit 2cbd36c with C1, C2, C3, Iwachu, and channel-asymmetry thresholds all fixed ex-ante.

FINDING 01

Recognition saturated: the ceiling in both cells

Phase A Recognition (C_P) distribution per cell

v0.19 audiophile headphones — 16 brands × 2 same-IL cells. C2 fails if C_P modal share ≥ 0.50.



Source: AIAS™ Presence Measurement Protocol v1.4 (SSRN 6799479) · v0.19-prereg-r1 · osf.io/ec6wh/v19/ Third System™ · thirdsystem.ai

Figure 1 · Phase A Recognition (C_P) score per brand, per cell. Each point is one brand; jitter spreads points within cells for visibility. Cell A_Heritage spans C_P = 2 (Koss) to 6 (Sennheiser, Beyerdynamic, Grado, Audio-Technica) with four of eight brands at the ceiling — C_P modal share = 0.500, exactly at the C2 strict-less-than boundary. Cell B_Boutique has seven of eight brands at C_P = 6 (only Spirit Torino at 5/6) — modal share = 0.875, well above the C2 threshold. Both cells fail C2; H_C3 routes to UNDETERMINED per pre-registered DEVIATIONS Rule 4. Cell A's failure-by-one-brand exposes a C2 operationalization edge case for AIAS™ 1.0 methodology refinement.

Cell A_Heritage produced a Recognition distribution that is substantively richer than v0.18's degenerate Cell A but operationally fails the pre-registered C2 threshold by a single brand at the ceiling. C_P scores: Sennheiser, Beyerdynamic, Grado, and Audio-Technica at 6/6; Stax at 5/6; Sony at 4/6; Denon at 3/6; Koss at 2/6. Five distinct C_P values across eight brands, range 2 to 6, mean ≈ 4.75. But the modal value (6/6) is held by four of eight brands — modal share = exactly 0.500. Under the pre-registered C2 rule (strict less-than 0.50), Cell A fails by the smallest possible margin.

Cell B_Boutique saturated. Seven of eight brands score C_P = 6/6 (Audeze, HiFiMan, Focal, Meze, ZMF Headphones, Final Audio, Dan Clark Audio); only Spirit Torino at 5/6 breaks the ceiling. Modal share = 0.875, well above the C2 threshold. This is the v0.18 ceiling pattern repeating — boutique audiophile brands are universally recognized by the panel as audiophile-headphone brands, leaving no within-cell Recognition variance for C3 ranking coherence to test.

Per DEVIATIONS Rule 4, both cells route H_C3 to UNDETERMINED. The verdict is methodologically appropriate: C3 measures rank correlation between Recognition and Recall, and when Recognition variance collapses the rank correlation is uninformative regardless of the Recall distribution. UNDETERMINED preserves the substantive interpretation: the substrate did not expose C3, not that C3 is false.

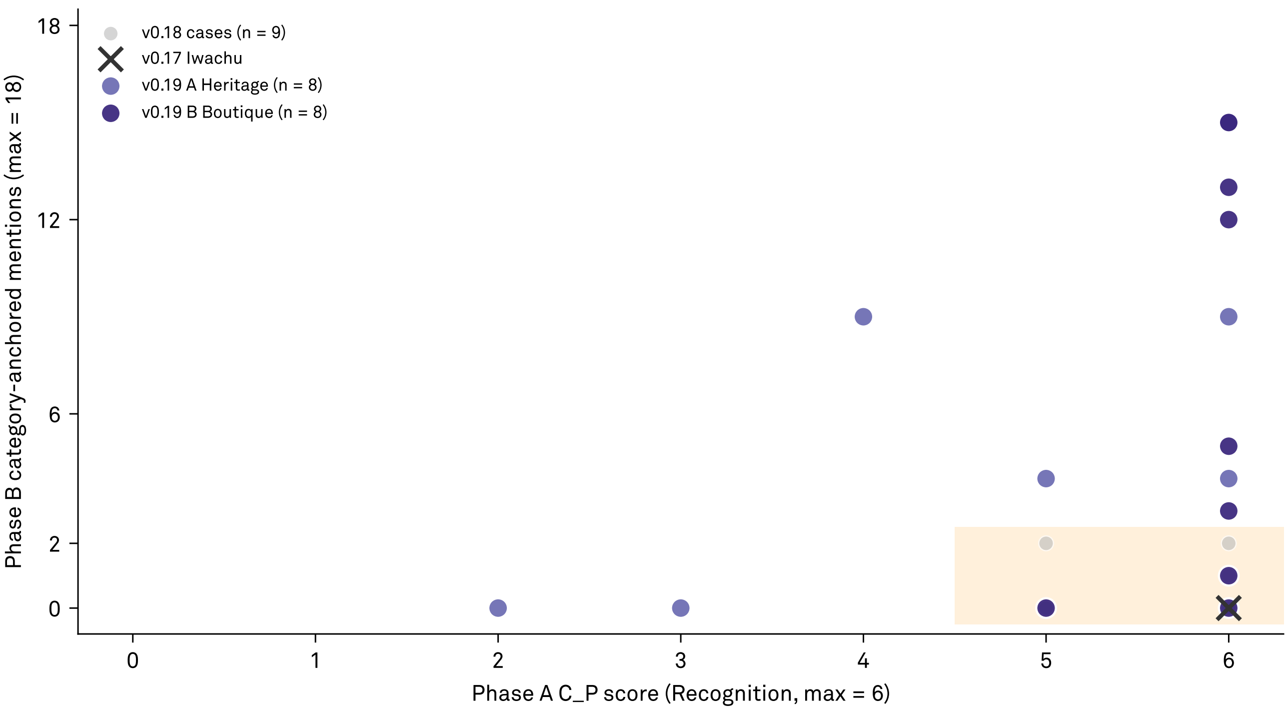
Cell A's failure at exactly 0.500 is a documented limit of the C2 operationalization, not a substrate failure per se. The distribution is qualitatively different from v0.18's Cell A (7/8 tied at 6/6, modal share 0.875). Five distinct values, range 4 across 8 brands — a substantively rich Recognition profile that the modal-share metric flags as uniform. AIAS™ 1.0's methodology layer should consider entropy-based or multi-statistic C2 specifications; the v0.19 boundary case is the empirical anchor motivating the refinement.

FINDING 02

Three more cases: dissociation lands in Boutique

Recognition × Recall dissociation — cumulative anchor base

Shaded quadrant: Iwachu-pattern (C_P ≥ 5 AND mentions ≤ 2). v0.19 contributes 3 cases to a base spanning 3 substrate families.



Source: AIAS™ Presence Measurement Protocol v1.4 (SSRN 6799479) · v0.19-prereg-r1 · osf.io/ec6wh/v19/ Third System™ · thirdsystem.ai

Figure 2 · Recognition × Recall dissociation scatter — cumulative anchor base across three substrate families. Each colored point is one v0.19 audiophile-headphone brand: Phase A C_P (x-axis, 0–6) versus Phase B category-anchored mention count (y-axis, 0–18). Gray ghost markers show v0.18 indie-fragrance Iwachu cases (n = 9); black × marks the v0.17 Japanese-cell Iwachu anchor. Shaded quadrant: Iwachu-pattern (C_P ≥ 5 ∧ mentions ≤ 2). v0.19 contributes 3 new cases (ZMF Headphones, Spirit Torino, Final Audio — all Cell B_Boutique). The AIAS™ v1.4 multi-component construct now spans three substrate families with 13 cumulative cases.

Three Iwachu-pattern cases on the audiophile substrate, all concentrated in Cell B_Boutique: ZMF Headphones (Phase A C_P = 6/6, Phase B category-anchored mentions = 1/18), Spirit Torino (C_P = 5/6, mentions = 0/18), and Final Audio (C_P = 6/6, mentions = 0/18). All three brands are panel-recognized as audiophile headphone brands at high consensus, yet the same panel does not surface them when asked to enumerate the category. The construct that v0.17 anchored and v0.18 generalized now extends to a third substrate family.

Cell A_Heritage contributes zero Iwachu-pattern cases — not by Recognition-floor design (as Cell C did in v0.18) but because Heritage brands that achieve C_P ≥ 5/6 (Sennheiser, Beyerdynamic, Grado, Audio-Technica, Stax) also surface substantively in Phase B category-anchored frames. The Heritage tier's broad LLM training-data depth supports both

Recognition and Recall; the Boutique tier's enthusiast-discourse presence supports Recognition without producing comparable Recall.

The cumulative anchor base for the v1.4 multi-component construct now spans three substrate families: 1 case from v0.17 (Iwachu, Japanese cell, cross-cultural), 9 cases from v0.18 (indie fragrance, English-language, IL-stratified), and 3 cases from v0.19 (audiophile electronics, English-language, brand-era stratified). Thirteen cases total. Before v0.18, the construct rested on one cross-cultural case; before v0.19, on a same-language substrate where the pattern could be argued to be IL-specific. v0.19 closes that argument: dissociation appears at the high-IL tier of fragrance and at the boutique tier of audiophile-electronics alike. The construct travels.

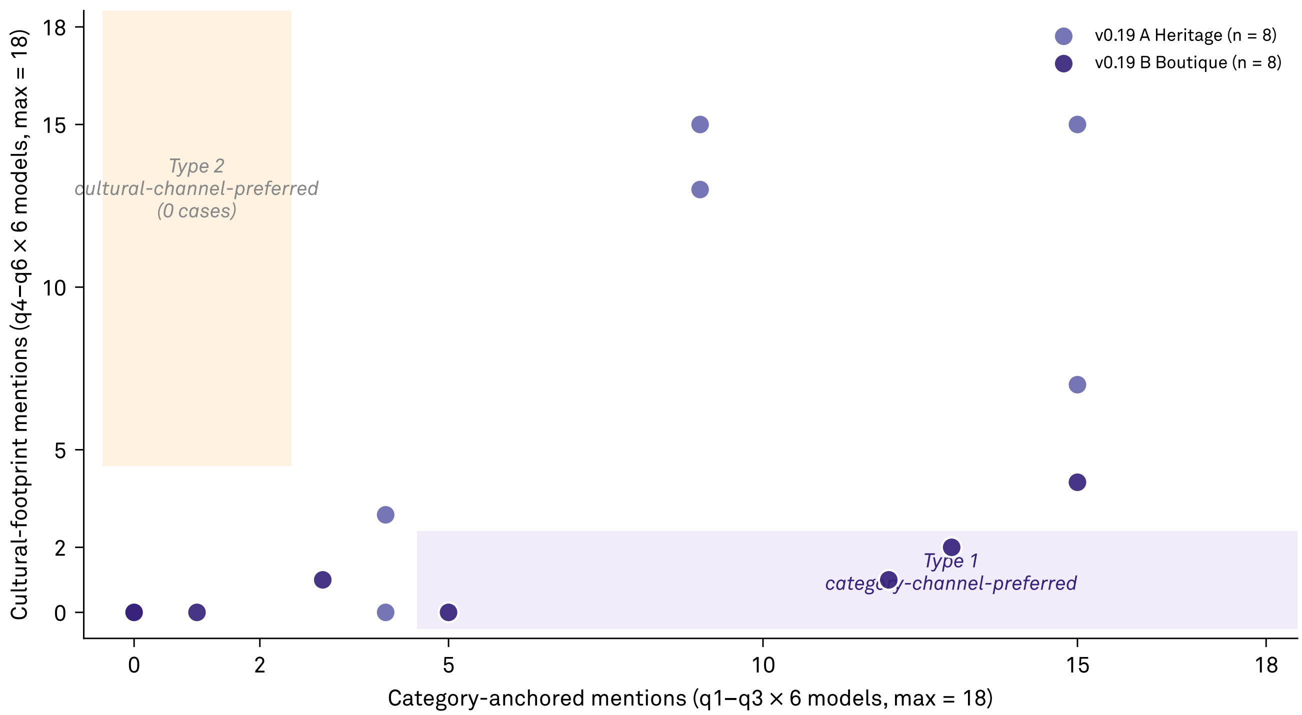
The cell-clustering pattern continues. v0.18 saw 6 of 9 cases in its highest-IL cell (75% of cell brands). v0.19 sees 3 of 3 cases in Boutique (37.5% of cell brands). Different cell-level prevalence, but the same directional story: dissociation concentrates in the cell whose brands operate further from the mainstream cultural-discourse surface. This is the empirical regularity the v1.4 construct predicts, observed twice on two distinct substrates.

FINDING 03

The channel that wasn't: cultural-footprint asymmetry

Channel asymmetry: category-anchored vs cultural-footprint Recall

Type 1: 3 v0.19 cases (Audeze, HiFiMan, Dan Clark Audio). Type 2: 0 cases — panel composition limit, not absent pathway.



Source: AIAS™ Presence Measurement Protocol v1.4 (SSRN 6799479) · v0.19-prereg-r1 · osf.io/ec6wh/v19/
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Figure 3 · Channel asymmetry scatter. Per-brand category-anchored mentions (x-axis, q1–q3 × 6 models = max 18) versus cultural-footprint mentions (y-axis, q4–q6 × 6 models = max 18). Lower-right shaded quadrant: Type 1 (category-channel-preferred; category-anchored ◧ 5 ▯ cultural-footprint ◧ 2). Upper-left quadrant: Type 2 (cultural-channel-preferred; category-anchored ◧ 2 ▯ cultural-footprint ◧ 5). Three Type 1 cases in v0.19: Audeze (13/18, 2/18), HiFiMan (12/18, 1/18), Dan Clark Audio (5/18, 0/18) — all Cell B_Boutique. Zero Type 2 cases. The Type 2 quadrant emptiness reflects panel composition (no mass-consumer brands in the Heritage × Boutique audiophile design), not the absence of the cultural-footprint Recall pathway.

v0.19's six-frame Phase B battery split category-anchored Recall (three frames testing audiophile-discourse retrieval) from cultural-footprint Recall (three frames testing mainstream consumer-discourse retrieval) so the two pathways could be observed independently. The v0.18 §4.3 sensitivity finding — Tom Ford and Chanel surfacing in category-anchored Phase B frames despite failing the niche-fragrance Recognition probe — predicted that v0.19 should produce **Type 2 cases**: brands with low category-anchored Recall but high cultural-footprint Recall.

Three Type 1 cases (category-channel-preferred): Audeze (category-anchored 13/18, cultural-footprint 2/18), HiFiMan (12/18, 1/18), and Dan Clark Audio (5/18, 0/18) — all Cell B_Boutique boutique audiophile brands. These brands surface strongly in audiophile discourse but minimally in mainstream cultural-recognition prompts, the expected profile for discourse-strong boutique brands.

Zero Type 2 cases — no brand in the v0.19 panel exhibits low category-anchored Recall paired with high cultural-footprint Recall. The cultural-footprint pathway is demonstrably operating (Phase B q4–q6 responses surface Bose, Beats, AirPods Max, Sony WH-1000XM, Skullcandy, JBL extensively) but the brands surfacing through it are not in the v0.19 panel. The Heritage × Boutique audiophile-electronics design places both cells inside the category-recognition envelope; nothing in panel tests the 'cultural without category' shape.

For AIAS™ 1.0's methodology layer, the Type 2 absence is a load-bearing finding: it identifies a panel-composition criterion v0.20+ must satisfy to test the cultural-footprint channel cleanly. A future phase needs a third cell of mass-consumer brands (Bose, Beats, Apple, Skullcandy, JBL) deliberately placed below the audiophile-category Recognition threshold. Type 2 cases will then be observable; the v1.5 methodology paper can formalize cultural-footprint Recall as a distinct Recall mode alongside category-anchored Recall.

Limitations

C2 operationalization at the strict-less-than boundary. Cell A_Heritage's C_P distribution [2, 3, 4, 5, 6, 6, 6, 6] is substantively richer than v0.18's degenerate Cell A (7/8 tied at C_P = 6/6, modal share 0.875). Five distinct C_P values across eight brands, range 4. But the modal-share metric captures only the most common value's prevalence and flags this as variance-degenerate at exactly 0.500. The strict-less-than C2 threshold is appropriately tight for catching v0.18-shape degeneracy but over-sensitive on mixed distributions where 50% of brands sit at one value with the other 50% distributed. AIAS™ 1.0 / v1.5 should consider entropy-based or multi-statistic C2 (e.g., distinct-value count + modal share, or Shannon entropy normalized to the cell).

Type 2 absence reflects panel composition, not pathway absence. The zero-Type-2 finding does not falsify the cultural-footprint Recall channel — Phase B q4–q6 responses surface Bose, Beats, AirPods Max, Sony WH-1000XM, Skullcandy, JBL extensively. The Heritage × Boutique audiophile-electronics design places both cells inside the category-recognition envelope, so no panel brand has the profile (low category-anchored Recall + high cultural-footprint Recall) the channel would produce. v0.20+ should include a mass-consumer cell deliberately placed below the audiophile-category Recognition floor.

Cross-cultural robustness analysis not applicable. Pre-registered DEVIATIONS Rule 6 requires Stax / HiFiMan exclusion analysis as Δ comparison against full-cell . Because both cells failed C2, was not computed for either cell, and the robustness analysis reports 'not applicable' for both. Cross-cultural confound exposure for v0.19 is therefore documented but unquantified. Future substrates with within-cell variance adequate to clear C2 will permit the robustness computation per Rule 6.

LLM panel temporal validity. The six-slot reference panel (claude-opus-4-5, claude-sonnet-4-5, gpt-4o, gpt-4o-mini, gemini-2.5-flash, gemini-2.5-flash-lite) reflects model versions in market as of mid-2026. Provider model substitutions over future-phase horizons are expected and documented in the DEVIATIONS protocol. v0.19 dissociation cases and channel-asymmetry pattern should be read as the result observed against the locked panel at acquisition (2026-05-20), not as permanent substrate properties.

What's next

Four pre-registerable directions emerge from the v0.19 verdicts. None requires retracting v0.19; each addresses a specific v0.19 finding.

Substrates with within-cell Recognition variance below the 0.50 modal-share floor. The C3 rescue did not land on the audiophile-electronics substrate. v0.20 should select a substrate where the within-cell Recognition distribution spans enough range that modal share sits cleanly below 0.50 in both cells — categories with a long tail of enthusiast-known but not universally-known brands (specialty board games, particular wine appellations, kitchenware tiers below premium) are candidates worth evaluating.

Mass-consumer cell inclusion to test Type 2 dissociation. v0.20 should include at least one cell of mass-consumer brands (in the headphone substrate: Bose, Beats, Apple AirPods, Skullcandy, JBL) deliberately placed below the audiophile-category Recognition floor. With brands in panel that the cultural-footprint frames surface but the category-anchored frames do not, Type 2 cases become observable and the cultural-footprint Recall channel can be formalized in v1.5.

v1.5 methodology paper: C2 operationalization refinement + cultural-footprint Recall mode. The methodology layer between v1.4 (current) and AIAS™ 1.0 (target) has two open items surfaced by v0.18 and v0.19 together. First, C2's modal-share-< 0.50 operationalization is over-sensitive on mixed distributions (v0.19 Cell A_Heritage) while under-sensitive on saturation (v0.19 Cell B_Boutique fails at 0.875). An entropy-based or distinct-value-count alternative is the candidate replacement. Second, cultural-footprint Recall surfaces in every multi-substrate phase but has no canonical methodology treatment; v1.5 can formalize it as a distinct Recall mode.

The C3 question remains open across the program. Four phases (v0.16, v0.17, v0.18, v0.19) have probed the C3 ranking-coherence test; none has produced a clean PASS on both cells of a substrate. The substrate-selection lever is the right one to pull next — v0.18's IL-gradient and v0.19's Heritage × Boutique designs both produced ceiling effects where the C3 test needs variance. The lever is not threshold loosening; it is substrate engineering. v0.20 carries that constraint into substrate selection from the start.

The three verdicts

Three pre-registered hypotheses; three verdicts of different shapes. H_C3 routes to UNDETERMINED per DEVIATIONS Rule 4 — the substrate did not expose the ranking-coherence test. H_Dissoc lands PARTIAL with three new Iwachu cases on a third substrate family. H_CulturalFootprint is descriptive only, returning a case-list that identifies a panel-composition limit for v0.20.

H	Pre-registered prediction	Result	Status
H_C3	C1 (n ≥ 12) → C2 (per-cell modal share < 0.50, both dimensions) → C3 (per-cell ≥ 0.50 AND bootstrap CI lower bound > 0.20, in ≥ 1 of 2 cells)	C1 (n = 16); C2 — Cell A_Heritage modal share = 0.500 (FAIL at boundary), Cell B_Boutique modal share = 0.875 (FAIL); C3 SKIPPED per Rule 4	UNDETERMINED
H_Dissoc	Iwachu pattern (C_P ≥ 5 category-anchored mentions ≤ 2) replicates on third substrate family (≥ 5 cases REPLICATED; 1–4 cases PARTIAL; 0 cases NOT_REPLICATED)	3 cases identified: ZMF Headphones (6/6, 1/18), Spirit Torino (5/6, 0/18), Final Audio (6/6, 0/18). All Cell B_Boutique. Cumulative anchor base = 13 cases across 3 substrate families.	PARTIAL
H_CulturalFootprint	Document Type 1 (category-preferred) and Type 2 (cultural-preferred) dissociation cases per cell. Descriptive output; no verdict.	Type 1: 3 cases (Audeze 13/18 vs 2/18; HiFiMan 12/18 vs 1/18; Dan Clark Audio 5/18 vs 0/18 — all Cell B). Type 2: 0 cases (panel-composition limit).	*DESCRIPTIVE*

Hypothesis details

The three verdicts in more detail, with the resolution path each took through the pre-registered decision cascade.

Resolved at C2 (variance adequacy). C1 panel adequacy clears cleanly (worldwide n = 16, per-cell n = 8, both above floors). C2 fails in both cells: Cell A_Heritage modal share = 0.500 (four of eight brands at C_P = 6/6), Cell B_Boutique modal share = 0.875 (seven of eight brands at C_P = 6/6). Per DEVIATIONS Rule 4, both cells route to UNDETERMINED. C3 is not computed — the substrate did not expose the test. The Cell A boundary case (failing by one brand at the ceiling) is documented as a methodological finding for AIAS™ 1.0 C2 operationalization refinement.

Three Iwachu-pattern cases identified on the v0.19 panel: ZMF Headphones (Phase A C_P = 6/6, Phase B category-anchored mentions = 1/18), Spirit Torino (5/6, 0/18), and Final Audio (6/6, 0/18). All three are Cell B_Boutique. PARTIAL routes between REPLICATED (≥ 5 cases) and NOT_REPLICATED (0 cases). Substantively this is the v0.19 headline: the v1.4 multi-component construct now spans three substrate families (v0.17 Japanese kitchenware + v0.18 indie fragrance + v0.19 audiophile electronics), with 13 cumulative cases across the three. The construct is no longer anchored by single-case or single-substrate evidence.

Three Type 1 cases (category-channel-preferred): Audeze, HiFiMan, Dan Clark Audio — all Cell B_Boutique brands with strong audiophile-discourse Recall and minimal mainstream cultural-footprint Recall. Zero Type 2 cases (cultural-channel-preferred). The Type 2 absence reflects panel composition: both cells of v0.19's Heritage × Boutique design sit inside the category-recognition envelope, so no panel brand has the 'low category-anchored Recall plus high cultural-footprint Recall' profile the cultural channel would produce. The descriptive finding identifies a v0.20 panel-composition criterion: include a mass-consumer cell below the category-recognition floor to test Type 2.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.4). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

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Underlying datasets

Pre-registration and DEVIATIONS log: osf.io/ec6wh/v19/PRE_REGISTRATION_v0_19.md and osf.io/ec6wh/v19/DEVIATIONS.md

Phase A and Phase B acquisition data: osf.io/ec6wh/v19/phase_a_results.csv and osf.io/ec6wh/v19/phase_b_results.csv

Scoring code, registry, figures: osf.io/ec6wh/v19/score_v0_19.py, panel_registry_v0_19.csv, thresholds_v0_19.json, scoring_output_v0_19.txt

Methodology log: thirdsystem.ai/methodology and osf.io/ec6wh.

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